任铎 2021111750

5-4 特征方程：$s^4+20Ks^3+5s^2+(10+K)s+1=0$

$\therefore$劳斯矩阵为

s^4	1	5	1
s^3	$20K$	$10+K$	0
s^2	$\dfrac{99K-10}{20K}$	1	0
s^1	$\dfrac{(10+K)(99K-10)-400K^2}{99K-10}$	0	0
s^0	1	0	0

特征方程所有系数都为正

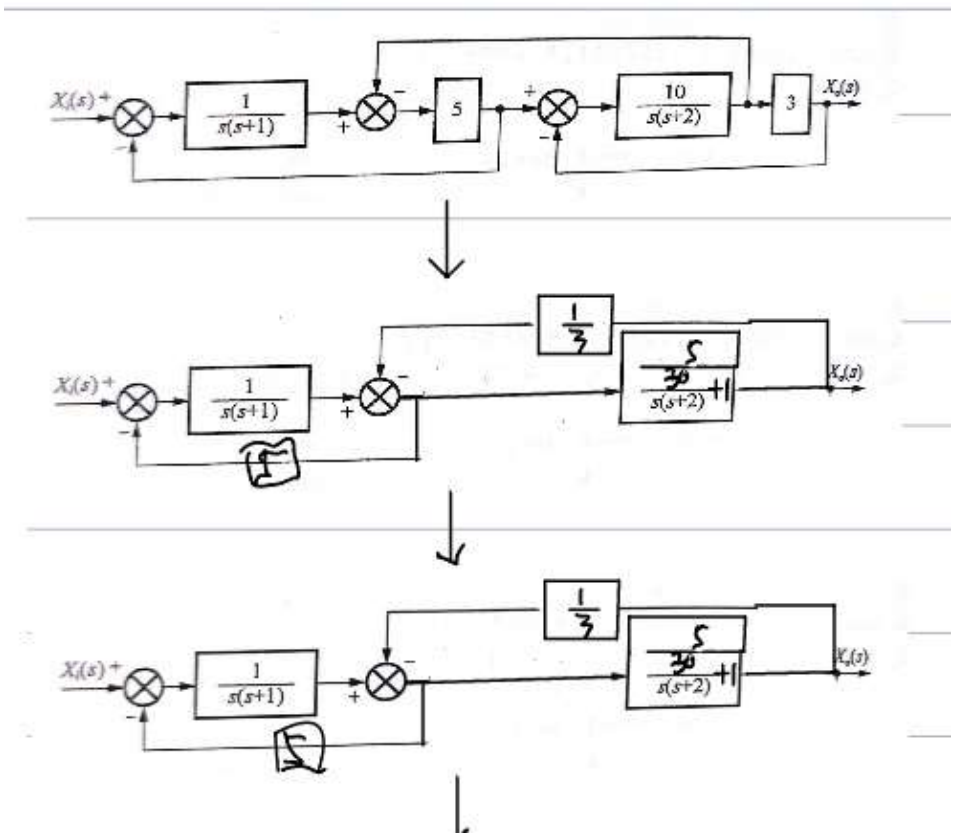
$\therefore 1,20K,5,10+K,1>0 \Rightarrow K>0$

系统稳定需劳斯表第一列全正

$\therefore 20K>0 \quad \dfrac{99K-10}{20K}>0 \quad (10+K)(99K-10)-400K^2>0 \quad \Rightarrow K>\dfrac{10}{99}$

综上$K>\dfrac{10}{99}$

5-6



5-8

开环传递函数$G_K(s)=\dfrac{K(s+5)(s+40)}{s^3(s+200)(s+1000)}$

$\therefore$特征方程是：$s^5+1200s^4+200000s^3+Ks^2+45Ks+200K=0$

劳斯表

s^5	1	200000	$45K$
s^4	1200	K	$200K$
s^3	$\dfrac{1200\times200000-K}{1200}$	$\dfrac{53800K}{1200}$	0
s^2	$\dfrac{K\cdot\dfrac{240000000-K}{1200}-1200\cdot\dfrac{53800K}{1200}}{\dfrac{240000000-K}{1200}}$	$200K$	0
s^1	$\cdots$	0	0
s^0	$200K$	0	0

“持续振荡”对应的是劳斯表首次某行首项为 **0**

$\therefore$令s^3行首项为 0则:$240000000-K=0\Rightarrow K=240000000$

此时辅助方程为s^4行$1200s^4+Ks^2+200K=0$代入K得$s^4+200000s^2+40000000=0$

令$s^2=-\omega^2$得$\omega^4-200000\omega^2+40000000=0\Rightarrow\omega=\sqrt{100000\pm\sqrt{9.6\times10^9}}\approx20\text{rad/s}$

5-9

图 5－22 系统，开环传递函数$G_K(s)=\dfrac{K(s+1)}{s^3+\alpha s^2+2s+1}$

$\Rightarrow$特征方程为$s^3+\alpha s^2+(2+K)s+(1+K)=0$

$\therefore$劳斯表

s^3	1	$2+K$
s^2	α	$1+K$
s^1	$\dfrac{\alpha(2+K)-(1+K)}{\alpha}$	0
s^0	$1+K$	0

令s^1行首项为 0：$\alpha(2+K)=1+K$

辅助方程$\alpha s^2+(1+K)=0$代入$s=j\omega$得$-\alpha\omega^2+1+K=0$令$\omega=2\therefore4\alpha=1+K$

联立方程:$\begin{cases}\alpha(2+K)=1+K\\4\alpha=1+K\end{cases}\Rightarrow\alpha=1，K=3$

5-10（2）